

CAPACITORS IN PHYSICS

Prepared by: Mahammad Mammadov, Fikrat Gurbanov, Murad Shirinov, Emil Rozenberg

Checked by: Telman Askeraliyev (Physics Teacher)

WHAT IS A CAPACITOR?

A capacitor is a fundamental passive electronic component designed to store electrical energy within an electric field. It achieves this energy storage by accumulating opposite electrical charges on two separate, closely positioned conductive plates that are completely isolated from one another by an insulating material known as a dielectric. Unlike chemical batteries, capacitors can discharge their stored electrical energy rapidly, making them indispensable for power conditioning, signal filtering, and high-frequency timing circuits.

CAPACITOR STRUCTURE & GOVERNING EQUATIONS

The basic configuration of a standard capacitor consists of two parallel conductive metal sheets (plates) facing each other. When a potential difference is established across these terminals, an equal magnitude of positive charge accumulates on one plate, while negative charge covers the opposite plate. The capability of this system to hold a charge per unit of electric potential is quantified as **Capacitance**.

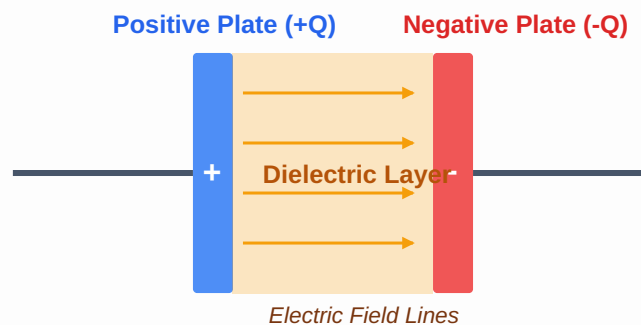


Figure 1: Internal structure layout showing charge distribution and electric field generation across a dielectric.

Capacitance Formula

$$C = Q / V$$

- **C**: Capacitance measured in *Farads (F)*
- **Q**: Accumulated Charge in *Coulombs (C)*
- **V**: Applied Potential Drop in *Volts (V)*

Stored Potential Energy

$$E = \frac{1}{2} C V^2$$

- **E**: Stored Electrostatic Energy in *Joules (J)*
- **C**: Component Capacitance in *Farads (F)*
- **V**: Terminal Working Voltage in *Volts (V)*

CHARGING DYNAMICS & FIELD BEHAVIOR

When connected to an external DC power source, electrons are actively pumped onto the negative terminal plate and drawn away from the positive terminal plate. This temporary current flow persists until the potential difference across the parallel plates reaches an equilibrium matching the exact voltage output of the driving source circuit. As charges securely settle on the boundaries, an electrostatic field lines up completely straight through the non-conductive dielectric material. This vector grid traps the energy safely, allowing it to remain completely intact even after the external supply circuit is fully isolated or disconnected.

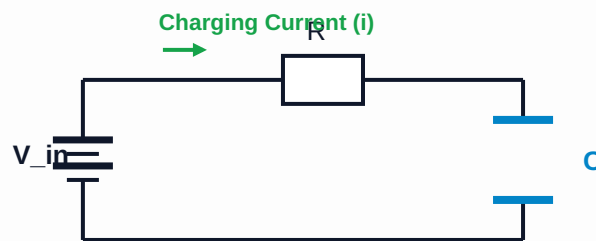


Figure 2: Basic loop schematic representing a charging series resistor-capacitor network.

SUMMARY: DECODING COMPONENT MARKINGS

In practical engineering workflows, tiny ceramic and film capacitors cannot fit full technical strings detailing their capacities. Instead, standard engineering systems utilize a concise three-digit numerical stamp to represent values explicitly in picofarads (\$pF\$), followed by an alphabetic tolerance character indicator.

Printed Code Stamp	Mathematical Decoding Base	Final Calculated Capacitance Value	Tolerance Threshold Rating
104K	$10 \times 10^4 \text{ pF} = 100,000 \text{ pF}$	100 nF (or 0.1 μF)	$\pm 10\%$ Operating Margin (K Indicator)
223J	$22 \times 10^3 \text{ pF} = 22,000 \text{ pF}$	22 nF (or 0.022 μF)	

Printed Code Stamp	Mathematical Decoding Base	Final Calculated Capacitance Value	Tolerance Threshold Rating
			±5% Operating Margin (J Indicator)
471M	$47 \times 10^1 \text{ pF} = 470 \text{ pF}$	470 pF	±20% Operating Margin (M Indicator)

** Note: The final trailing numerical integer always commands the scaling power-of-ten exponent multiplier, defining exactly how many trailing zeros must append to the preliminary two-digit baseline.*